A non-virtually-degenerate p-polygonal equality in ℓ_p^2 for every $p > 2$

Autonomous counterexample for arXiv:1203.5837

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Abstract

Kelleher–Miller–Osborn–Weston proved that virtual degeneracy is sufficient for a nontrivial p-polygonal equality in every L_p space and asked whether it is necessary when $p > 2$. It is not. For every $p > 2$ we give a three-vertex weighted simplex in ℓ_p^2 whose two scalar-coordinate gaps have opposite signs and cancel after one explicit rescaling. One coordinate simplex is nondegenerate, so the equality is not virtually degenerate.

1 Gap convention and result

For a normalized signed simplex

$$D = [x_i(m_i); y_j(n_j)]_{s,t}, \qquad m_i, n_j > 0, \quad \sum_i m_i = \sum_j n_j = 1,$$

put

$$\Gamma_p(D) = \sum_{i,j} m_i n_j \|x_i - y_j\|_p^p - \sum_{i<i'} m_i m_{i'} \|x_i - x_{i'}\|_p^p - \sum_{j<j'} n_j n_{j'} \|y_j - y_{j'}\|_p^p.$$

A nondegenerate simplex with $\Gamma_p(D) = 0$ is a nontrivial p-polygonal equality in the terminology of the source.

Theorem 1. *For every $p > 2$, virtual degeneracy is not necessary for a nontrivial p-polygonal equality in an L_p space. More precisely, ℓ_p^2 contains a completely refined signed $(2,1)$-simplex D such that $\Gamma_p(D) = 0$ but D is not virtually degenerate.*

2 Construction and proof

Fix $p > 2$ and define $c = c_p > 0$ by

$$c^p = \frac{2^{p-2} - 1}{2^{p-1} + \frac{1}{4}}. \tag{1}$$

The numerator is positive precisely because $p > 2$. In ℓ_p^2 set

$$x_1 = (-1, 0), \qquad x_2 = (1, c), \qquad y_1 = (0, 2c), \qquad m_1 = m_2 = \frac{1}{2}, \quad n_1 = 1. \tag{2}$$

All three vertices are distinct and all weights are positive, so $D = [x_1(1/2), x_2(1/2); y_1(1)]$ is completely refined and in particular nondegenerate.

There is only one within-side term. Directly from (2),

$$\begin{aligned}\Gamma_p(D) &= \frac{1}{2}\bigl(1+(2c)^p\bigr) + \frac{1}{2}\bigl(1+c^p\bigr) - \frac{1}{4}\bigl(2^p+c^p\bigr)\\ &= 1 - 2^{p-2} + \left(2^{p-1}+\frac{1}{4}\right)c^p = 0,\end{aligned}$$

where the last equality is (1). Thus D gives a nontrivial p-polygonal equality.

It remains only to exclude virtual degeneracy. In the first scalar coordinate the induced signed simplex is

$$[-1(1/2),,1(1/2);,0(1)].$$

Its three vertices are distinct. At the scalar point -1, for example, the total weight on the x-side is $1/2$ and that on the y-side is zero. Hence this scalar simplex is nondegenerate. Virtual degeneracy in ℓ_p^2 requires degeneracy in every scalar coordinate, so D is not virtually degenerate. This proves Theorem 1. $\square$

3 Mechanism and scope

The construction is a cancellation that is unavailable below exponent two. The first coordinate contributes

$$1 - 2^{p-2} < 0$$

to the gap. Before scaling, the second coordinate—the scalar simplex $[0(1/2), 1(1/2); 2(1)]$—contributes

$$2^{p-1} + \frac{1}{4} > 0.$$

Because $\|z\|_p^p$ is additive over coordinates and scaling a coordinate by c multiplies its gap by c^p, equation (1) cancels the two contributions exactly.

This answers only the explicit necessity question in the first paragraph of the source's open-problems section (source PDF page 17). It does not classify all p-polygonal equalities for $p > 2$. The later arXiv:2404.06658 proves a general existence criterion and settles the source's separate Schatten-class existence question, but does not discuss virtual degeneracy; the elementary counterexample above is independent of that result.